SAL-04 · SALARY AND SKILLS REPORT

The skills demand taxonomy

A controlled vocabulary for skills, and an honest definition of what "demand" can mean

VERSION 1.0 · DRAFT
2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.

AI proposes. The professional disposes.

Summary

The structured vocabulary the Institute's salary and skills report uses to classify skills across five facets — technical method, tool category, domain, data and AI, and behavioural — with a permanent code for each, the rules for coding free text into it, and a precise definition of what the report may call "demand". The short version of that definition: what respondents were asked for is a reported requirement, not market demand, and the report says so in those words.

About this document

Document	SAL-04
Series	Salary and Skills Report
Version	1.0
Status	draft
Date	2026-08-04
Unresolved placeholders	3

Contents

The skills demand taxonomy

In one paragraph. This document defines the structured vocabulary the Institute's salary and skills report uses to classify skills across five facets — technical method, tool category, domain, data and AI, and behavioural — with a permanent code for each, the rules for coding free text into it, and a precise definition of what the report may call "demand". The short version of that definition: what respondents were asked for is a reported requirement, not market demand, and the report says so in those words.

Who this is for. Analysts coding survey responses; heads of function and capability leads who need a skills vocabulary that does not change every time a tool is renamed; and anyone assessing whether a published skills statistic means what its headline says.

1. What goes wrong when skills are counted without a vocabulary

Ask a hundred practitioners what skills their job requires and you get a hundred phrasings of perhaps thirty things. "Forecasting", "EAC", "estimate at completion", "predicting the outturn" and "cost projection" are one skill written five ways. "Data skills" is between four and nine different skills depending on who wrote it. "Excel" is a tool, a proxy for modelling capability, and sometimes a complaint.

Left uncoded, that free text produces a ranked list whose ordering is an artefact of vocabulary rather than of anything in the world — the skill with the most common single phrasing wins. Coded carelessly, it produces something worse: a list that looks authoritative and is unreproducible, because the person who made the coding decisions did not write them down.

A controlled vocabulary fixes both, and costs something in return. It cannot represent a skill nobody anticipated, and it imposes boundaries that will occasionally be wrong. The compensating controls are in §5: unmatched text is coded as unmatched and reported rather than forced, and the codebook has a defined review point where new codes are added — deliberately not continuously, because a vocabulary that changes mid-cycle cannot be compared with itself.

2. Five facets

Skills are classified on five facets. A facet is not a level of importance; it is a kind of skill, and each kind behaves differently over time and needs different treatment in analysis.

Facet	Code	What it holds	Why it is separate
Technical method	`T`	The discipline itself — how the work is done, independent of any tool	The slowest-moving facet, and the one the Body of Knowledge maps to directly
Tool category	`L`	Categories of software and platform, never named products	Products churn; categories persist. Naming products would date within a year and function as endorsement
Domain	`D`	Sector, contracting model and standards context	Portable-looking skills are often domain-bound, and this facet is what reveals it
Data and AI	`A`	Data handling, analysis, automation and the responsible use of AI tooling	The fastest-moving facet; kept separate so its volatility does not contaminate the others
Behavioural	`B`	Professional behaviour and judgement	Measurable only through anchored self-report, and reported with heavier caveats than the rest

Code structure: `FACET-GROUP-SKILL`, for example `T-EVM-EAC`. Codes are permanent. A retired code is never reused for a different skill, for the same reason a document ID is never reused: it is a citation key, and anyone comparing two cycles must be able to trust that the same code means the same thing.

3. The vocabulary

Each entry gives the code, the skill, what it means in one line, and what evidence of it looks like — the evidence column is what makes the code applicable by two different analysts to the same free text.

3.1 Facet T — technical method

Code	Skill	What it means	What evidence looks like	BoK
T-PLN-DEV	Schedule development	Building a schedule that represents how the work will actually be done	A network with defensible logic and a written schedule basis	10
T-PLN-INT	Network and logic integrity	Keeping the schedule's logic sound as it changes	Diagnosed and corrected logic defects; documented open ends and constraints	10
T-PLN-PRG	Progress measurement	Establishing how progress is claimed and verified	Rules of credit applied consistently across a scope	6
T-PLN-CPA	Critical path and float analysis	Reading what the network says about completion and slack	Float analysis that changed a decision	10
T-PLN-TIA	Time impact analysis	Isolating the schedule effect of a discrete event	A modelled impact that survived challenge	10, 7
T-CST-CBS	Cost structure design	Designing a cost breakdown that aligns to scope and to the accounts	A cost breakdown structure mapped to both the work breakdown structure and the code of accounts	5
T-CST-BUD	Budgeting and control accounts	Distributing a budget into controllable units	Control accounts with named owners	3, 5
T-CST-ACT	Commitment and actual capture	Getting real cost into the report completely and on time	Committed, incurred and paid tracked distinctly	5, 11
T-CST-ACC	Accruals and cut-off	Reporting the period's cost in the period	An accrual schedule that reconciles to finance	2, 11
T-CST-VAR	Variance analysis	Explaining a difference in terms someone can act on	Variance narrative naming cause, not restating the number	4, 5
T-EST-QTO	Quantity take-off	Deriving quantities from a design of stated maturity	Take-off with a stated source and maturity	3
T-EST-MTH	Estimate method and class	Choosing a method appropriate to definition and stating its class	An estimate labelled by class with an accuracy range stated as a range concept	3
T-EST-BEN	Benchmarking and norms	Using comparable data critically, including rejecting it	A norm applied with a documented adjustment	3
T-EST-BOE	Basis of estimate	Writing down what the estimate assumes, includes and excludes	A basis document a stranger could audit	3
T-EVM-TEC	Earned value technique selection	Choosing a measurement method suited to the work type	Technique chosen per work package, not per project	6
T-EVM-IND	Performance indices	Computing and, more importantly, interpreting indices	An index read alongside its data quality, not in isolation	6
T-FCT-EAC	Estimate at completion	Selecting, computing and defending a forecast method	Multiple methods compared and one chosen with a reason	6, 3
T-FCT-CSH	Cash flow forecasting	Forecasting the timing of money, not only its total	A profile reconciled to schedule and to commercial terms	3, 7
	Risk identification			12

Code	Skill	What it means	What evidence looks like	BoK
T-RSK-IDN		Surfacing what could happen, in usable form	Risks written as cause-event-effect, not as concerns	
T-RSK-QNT	Quantitative risk analysis	Modelling uncertainty in cost or schedule	A simulation with defensible ranges and correlation treatment	12
T-RSK-CTG	Contingency derivation and drawdown	Deriving contingency from analysis and controlling its release	Contingency traced to risks and drawn down against them	12, 3
T-COM-CHG	Change identification and valuation	Recognising a change and pricing it	A change register where the entry date matches the event	7
T-COM-ENT	Entitlement and notice	Managing the records that make a position defensible	Notices issued in time, with contemporaneous records	7
T-COM-CLM	Claims and extension of time	Constructing a substantiated narrative from records	A claim narrative supported by its own evidence trail	7
T-GOV-BLC	Baseline control	Keeping the baseline meaningful under change	A change-controlled baseline with a traceable history	4, 8
T-GOV-PCE	Controls execution planning	Writing down how controls will be run before running it	A project controls execution plan in use, not filed	8
T-GOV-ASR	Assurance and health check	Testing whether reported performance is credible	A health check that produced corrective actions	4
T-RPT-MGT	Management reporting	Producing reporting that changes a decision	A report whose recommendations were acted on	4

3.2 Facet L — tool category

Code	Category	Scope of the code
L-SCH	Planning and scheduling software	Any critical-path scheduling platform
L-CST	Cost management or control system	Dedicated project cost control platforms
L-ERP	Enterprise resource planning system	Finance and procurement systems as they touch project cost
L-RSK	Risk analysis and simulation tools	Quantitative cost and schedule risk platforms
L-BI	Business intelligence and dashboarding	Reporting and visualisation platforms
L-SPR	Spreadsheet as a primary control tool	Including structured modelling, not incidental use
L-DOC	Document control and correspondence systems	Including the controls-relevant record trail
L-COD	Programming or scripting	Any language used to manipulate controls data
L-AIT	AI assistant or large language model tooling	General-purpose and embedded assistants
L-INT	System integration and interfacing	Moving data between the above without losing its meaning

Named products are never published. They are collected as free text at Q42 (`SAL-02`) and used only to assign the category. Publishing a ranked list of products would function as an endorsement the Institute has no basis to give, would age within a year, and would distract from the finding that matters: which categories of capability a role is expected to hold. [CONFIRM: whether an annex lists the named products encountered, in alphabetical order with no counts and an explicit non-endorsement statement, or whether product names are discarded entirely after coding]

3.3 Facet D — domain

Code	Domain skill	What it means
D-SEC-HVY	Heavy industrial and energy delivery	Controls practice under the constraints of large capital plant and site delivery
D-SEC-INF	Infrastructure and transport	Long linear or networked assets, often publicly scrutinised
D-SEC-BLD	Buildings and property	Trade-package structures and shorter cycles
D-SEC-REG	Highly regulated environments	Nuclear, pharmaceutical, defence and similar: qualification and evidence regimes that change how controls is done
D-SEC-CHG	Technology and business-change portfolios	Iterative delivery where scope is discovered rather than specified
D-CON-LSM	Lump sum and fixed price	Controls under fixed-price commercial pressure
D-CON-REI	Reimbursable and target cost	Open-book regimes, pain-gain mechanisms, client audit
D-CON-ALL	Alliance and integrated delivery	Shared-risk arrangements and their reporting demands
D-CON-PPP	Concessions and public-private arrangements	Long-horizon financing structures as they constrain controls
D-STD-EVM	Earned value guidance and its conventions	Working to a recognised earned value management framework
D-STD-RSK	Risk management standards	Applying recognised risk principles, such as those in ISO 31000
D-STD-TCM	Total cost management practice	Working within the AACE International total cost management body of practice
D-STD-PMB	Project management frameworks	PMBOK and comparable frameworks as they meet controls
D-STD-AGL	Adaptive delivery frameworks	Scrum and comparable approaches, and their controls implications

3.4 Facet A — data and AI

Code	Skill	What it means	BoK
A-DQL	Data quality and structure	Knowing whether the data can bear the analysis about to be done to it	13, 4
A-MDL	Data modelling for controls	Structuring controls data so cost, schedule and change can be joined	13
A-ANL	Analysis and applied statistics	Trend, distribution and comparison, used correctly	13, 4
A-VIS	Visualisation and metric definition	Defining a metric precisely and displaying it without distorting it	4
A-AUT	Automation of recurring work	Removing manual steps from a reporting cycle reliably	13
A-PRD	Predictive and machine-assisted forecasting	Using algorithmic methods to support a forecast a human still owns	13, 6
A-PRM	Task framing for AI tools	Specifying a task to an AI tool precisely enough to get usable output	13
A-VAL	Validation of AI output	Checking what a tool produced before anyone relies on it	13
A-EXP	Explainability and bias awareness	Understanding what a model did and where it can be systematically wrong	13
A-GOV	AI governance and permissible use	Knowing what is allowed, recording what was used, and keeping the audit trail	13

A-VAL and A-GOV carry the Institute's position directly: AI proposes; the professional disposes. They are separate codes rather than an aspect of the others because they are separately absent — a respondent can be fluent with the tools and have no validation practice at all, and that combination is precisely what a skills report should be able to see.

3.5 Facet B — behavioural

Code	Skill	What it means
B-COM	Explaining a number to a non-specialist	Making a forecast or variance land with someone who does not do this job
B-POS	Holding a position under pressure	Maintaining a defensible number when it is unwelcome
B-STK	Stakeholder management	Working across client, contractor and function boundaries
B-COL	Cross-discipline collaboration	Working effectively with delivery, commercial and finance
B-JUD	Judgement under incomplete data	Deciding when the data will not settle it, and saying what is assumed
B-LRN	Method improvement	Improving how the work is done, not only doing it
B-DEV	Developing others	Building capability in a team
B-ETH	Professional conduct	Recognising and escalating a reporting integrity problem

Behavioural codes are reported with the heaviest caveats in the report, because they rest entirely on anchored self-report. `SAL-02` Q40 uses behavioural anchors precisely to make this facet less self-flattering, and the limitation is stated wherever a behavioural finding appears.

4. Skill entries — what a code carries

Every code in the master codebook carries seven fields, and a code without all seven is not fielded: the code itself; the facet; the display label used in the instrument; the one-line definition; the evidence descriptor; the mapped Body of Knowledge domain or domains; and the cycle in which the code was introduced. Retired codes additionally carry the cycle of retirement and the reason.

5. Coding free text into the vocabulary

Free text arrives from three places: the "other" fields at Q07 and Q25, the product names at Q42, and the open question at Q52.

The procedure.

1. **Screen for identifying detail first.** Employer names, project names, colleagues' names and anything else identifying is removed — removed, not paraphrased — before any coding begins.
2. **Code to the vocabulary, or to `UNC`.** A segment of text is assigned one or more codes. Where no code fits, it is coded `UNC` (unmatched) with the text retained. `UNC` is reported as a count. It is never distributed across neighbouring codes to make a table look complete.
3. **Code segments, not responses.** One open answer may carry three skills; forcing one code per respondent silently discards two-thirds of what was said.
4. **Double-code independently.** A defined share of the free text is coded by two analysts working separately, and their agreement is measured with a named statistic — Cohen's kappa for two coders on nominal codes, or Krippendorff's alpha where more than two coders or partial agreement are involved. The statistic and the minimum acceptable value are fixed before coding begins [CONFIRM: which agreement statistic is used and the minimum acceptable value below which coding is redone rather than reconciled].
5. **Resolve disagreements by rule, not by seniority.** A third analyst adjudicates; the adjudication is recorded; if the same boundary is adjudicated repeatedly, that is a codebook defect and goes to the review point rather than being absorbed by the analysts.
6. **Do not add codes mid-cycle.** New codes are added only at the codebook review, between cycles. A vocabulary that grows during coding cannot be compared with itself, and the first responses coded were coded under a different scheme from the last.
7. **Publish the codebook version** used for each cycle, with the change log [CONFIRM: codebook owner, version numbering scheme and the review point at which new codes may be introduced].

6. What "demand" may mean here

This is the section that decides whether the skills half of the report is research or decoration. The instrument measures three distinct constructs, and the report never merges them or lets one borrow the other's name.

Construct 1 — reported requirement. From Q41: a respondent states that a skill was explicitly required of them, in a documented form (job specification, objective, appraisal or interview), within the twelve months to the reference date. What it measures: what these respondents were asked for. What it does not measure: what employers who did not respond want, what unfilled vacancies specify, or what anyone will want next year.

Construct 2 — held capability. From Q40: a respondent's anchored self-assessment of what they can do. What it measures: claimed capability among respondents. What it does not measure: assessed competence — that requires assessment, which is what the Institute's certification does and this survey does not.

Construct 3 — capability gap. Computed within a respondent: a skill explicitly required at Q41 and held at the two lowest anchors at Q40. Computed per person and then counted, never inferred by comparing two aggregate distributions — comparing a required-skills distribution with a held-skills distribution across different people produces a "gap" that may not exist for any individual.

Economists mean something specific by demand: a schedule of quantity against price, evidenced by vacancies, time-to-fill and wage movement. The Institute's survey collects none of that. It therefore never uses the word without qualification.

Never written	Written instead
"The most in-demand skill in project controls"	"The skill most often reported as explicitly required by respondents in this cell"
"Demand for X grew"	"A larger share of respondents reported X as explicitly required than in the previous cycle" — and only where §7 permits a comparison at all
"There is a shortage of X"	"A capability gap for X was reported by respondents in this cell"
"Employers want X"	"Respondents reported that X was required of them"
"X is the future of the profession"	Nothing. This is not a measurement

Every one of the permitted forms is longer and duller than the phrase it replaces. That is the cost of saying only what was measured, and it is the whole reason to read the report rather than a headline about it.

7. Reporting rules for skills findings

1. **Cell thresholds apply identically.** The minimum response counts and suppression rules in `SAL-01` §6 govern skills findings exactly as they govern pay findings. A skills table is not exempt because it contains no money.
2. **Item-level denominators.** Skills matrices lose respondents partway through. Every skills figure carries the count of respondents who answered that item, not the survey total.
3. **Ranks are bands.** Where the difference between adjacent skills is within the reporting precision, they are shown as a tied band. A ranked list with no tie rule invites readers to treat noise as order.
4. **No top-ten lists without the whole list.** Publishing only the top of a ranking hides how flat the ranking is. The full ordering is published in an annex whenever any part of it is quoted.

5. **No cross-cycle skills comparison** unless the item wording, the anchors and the code were all unchanged, and both cycles meet the cell threshold. Adding a skill to the vocabulary resets that skill's series to its first cycle.
6. **Facet A findings carry a currency warning.** The data and AI facet moves fastest; a finding from it is labelled with its reference period wherever it appears, because a twelve-month-old statement about AI tool use is a historical statement.
7. **Behavioural findings carry the self-report caveat** in the same paragraph, not in a footnote.

8. How this goes wrong

The vocabulary becomes a wish list. Someone adds the skills the Institute would like the profession to value. The vocabulary must describe what exists, including the parts nobody is proud of — the spreadsheet category exists for exactly that reason.

Tool codes swallow method codes. A respondent who names a scheduling product is coded as having schedule development capability. They are different things, and the facet separation exists to keep them apart.

Coding drift. Over a long coding run, a boundary case is decided one way early and the other way late. Double-coding a sample at the start, middle and end of the run catches it; a single agreement check at the start does not.

`UNC` gets tidied away. Unmatched text is the taxonomy's error signal. Distributing it into the nearest codes destroys the only evidence that the vocabulary is incomplete.

The gap that exists only in aggregate. Two distributions are compared, a difference appears, and it is reported as a skills gap. Construct 3 is computed per respondent for this reason.

"Demand" leaks back in. The report is careful and the promotional post is not. The language table in §6 applies to the headline, the social copy and the conference slide, not only to the body text.

9. Checklist — before a skills finding is published

- [ ] Every code used is in the published codebook version for this cycle
- [ ] Identifying detail removed from free text before coding, not after
- [ ] Double-coded sample drawn from across the whole coding run, and agreement reported
- [ ] `UNC` count reported, and no `UNC` text redistributed
- [ ] No code added mid-cycle
- [ ] Item-level response count shown on every skills figure
- [ ] Ranked lists carry a tie rule; nothing quoted from a ranking without the full ordering available
- [ ] Construct named explicitly — reported requirement, held capability or capability gap
- [ ] The word "demand" checked against the language table in §6, in the report and in every derived post
- [ ] Facet A findings labelled with their reference period; facet B findings carry the self-report caveat
- [ ] No product name published in any ranked or counted form

Related

- `SAL-02 — The survey instrument` — the skills matrices, tool question and open question this vocabulary codes
- `SAL-03 — Role taxonomy and levelling` — the roles these skills are reported against
- `SAL-05 — Report template and data tables` — the skills tables and their standing caveats
- `CMP-08 — Data, digital and AI competencies in depth` — what competence in the data and AI facet actually requires
- `AIG-12 — The AI-literate controls professional` — the Institute's position on the capability facet A measures

Sources and standards

No external source is cited. Standards named in facet D — ISO 31000 risk management principles, the AACE International total cost management practice, PMBOK, the Scrum Guide, and recognised earned value management guidance — are named as objects of the domain codes and are not reproduced, summarised or represented here; where the report describes them it does so in our own words. Body of Knowledge domain numbers refer to the thirteen domains in `docs/bok/`. The inter-coder agreement statistics named in §5, Cohen's kappa and Krippendorff's alpha, are standard content-analysis measures described here in our own words.

Status and version

Founding-stage document · Version 1.0 · draft · Reviewed under PCI governance. PCI makes no claims of accreditation or recognition beyond what is true today.